CYBERSECURITY CHALLENGES IN DECENTRALIZED TECHNOLOGIES

I. INTRODUCTION TO DECENTRALISED TECHNOLOGIES

Decentralised technologies are systems where control and decision-making are distributed across many participants instead of a single central authority like a bank, company, or government, this means using a distributed network of computers (nodes) that all maintain and verify a shared database (a ledger). This includes Blockchain networks like Bitcoin and Ethereum, Decentralised Finance (DeFi) platforms, Non-Fungible Tokens (NFT's) and Decentralised Applications (dApps). Their key properties are transparency, immutability and reduces dependence on other middlemen or authorities.

II. HOW BLOCKCHAIN-BASED SYSTEMS OPERATE

A blockchain records transactions in order in blocks where each block is linked to the previous block using cryptographic hash. Each block contains transaction data, timestamp and cryptographic hash of the previous block. The linking of the blocks makes the tampering difficult.

Participants (called nodes) each hold a full copy of the ledger. When a new transaction is proposed, instead of trusting one server, users trust the network to verify transactions through **consensus mechanism**:

<u>Proof of Work (PoW)</u> — used by Bitcoin (miners solve complex computational problems)

<u>Proof of Stake (PoS)</u> — used by Ethereum (validators stake cryptocurrency)

Smart contracts are self-executing code stored on the blockchain. Once deployed, they are usually immutable.

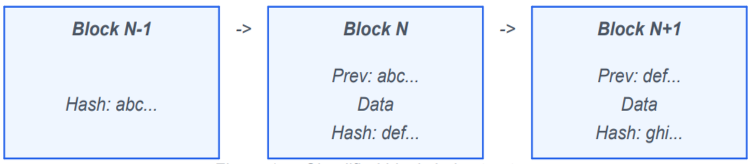
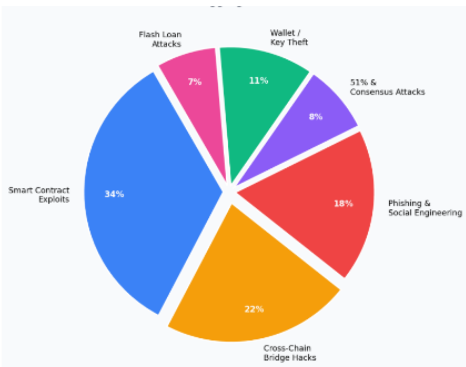


Figure 1 — Simplified blockchain structure

III. COMMON CYBERSECURITY THREATS

Decentralized systems face many threats just like the traditional systems, but the impact is often worse because stolen assets are hard to recover and transactions are usually final. Major risks include smart contract bugs, private key theft, phishing, manipulation, consensus attacks, governance abuse, and cross-chain bridge exploits contract bugs, private key theft,



phishing, manipulation, 51% and consensus attacks, governance abuse, and cross-chain bridge exploits.

IV. SMART CONTRACT VULNERABILITIES AND ATTACKS

Smart contracts are self-executing programs stored on a blockchain. They run automatically when certain conditions are met. But if there is a bug in the code, attackers can exploit it — and once deployed, smart contracts are usually very difficult to fix.

- REENTRANCY ATTACK
  This is one of the most famous vulnerability types in blockchain history. The idea is straightforward, a malicious contract repeatedly calls back into a vulnerable contract before it finishes executing, allowing repeated withdrawals before balances are updated; the 2016 DAO hack is the best-known case linked to this issue.
  In DAO hack of 2016, hackers exploited reentrancy to steal approximately $50–60 million in ETH. This event led to creation of Ethereum and Ethereum Classic.
- FLASH LOAN ATTACKS
  Flash loans are uncollateralized loans that must be borrowed and repaid within a single blockchain transaction.
- OTHER SMART CONTRACT VULNERABILITIES
  Integer overflow/underflow — If a number exceeds the maximum value a variable can hold, it 'wraps around' to zero or a negative value
  Access Control issues — functions that should be private being accidentally left public.

V. 51% ATTACKS AND THEIR IMPACT

A 51% attack occurs when a single entity gains control of more than half of a blockchain network's mining power (in PoW) or staked tokens (in PoS). This can allow the attacker to reorganize recent blocks, censor transactions, and attempt double spending, which weakens confidence in the network even if the attacker cannot simply rewrite the entire history. Ethereum Classic (ETC) suffered multiple 51% attacks in 2020.

VI. WALLET THEFT, PHISHING, AND SOCIAL ENGINEERING RISKS

Blockchain systems may be decentralized, but users still depend heavily on wallets and private keys. If a private key or seed phrase is stolen, control over the assets is usually lost immediately, which makes phishing and social engineering some of the most practical attacks in Web3. Common scams include fake wallet websites, malicious browser pop-ups, phishing links sent on social media, and fraudulent support messages asking users to reveal recovery phrases. Because transactions are signed by the user, the blockchain itself may treat a scam transaction as valid even when it was socially engineered.

VII. SECURITY CHALLENGES IN DEFI PLATFORMS

DeFi (Decentralized Finance) aims to rebuild traditional financial services — such as lending, borrowing, and trading — on blockchain technology, without needing banks. It has grown enormously in value and popularity, but it remains one of the most heavily attacked areas in the crypto space. They depend heavily on smart contracts. Major DeFi security challenges include:

- Flash loan attacks are a major DeFi risk because they let attackers borrow large amounts of funds in one transaction and use that temporary capital to manipulate prices or governance.

- Oracle manipulation — DeFi protocols rely on price oracles to know asset values. If an attacker manipulates the oracle feed, they can exploit lending and trading protocols for profit.
- Smart Contract bugs, Liquidity pool exploits and Lack of regulation

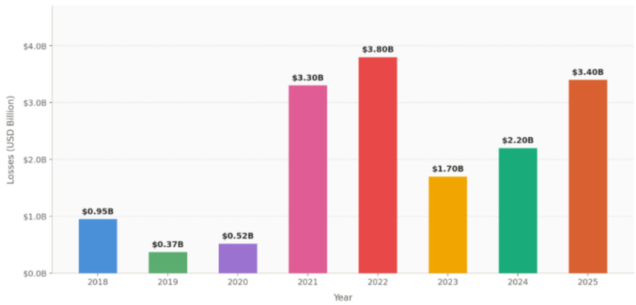
VIII. CROSS-CHAIN BRIDGE VULNERABILITIES AND REAL-WORLD INCIDENTS

Cross-chain bridges allow users to move assets between different blockchains, for example from Ethereum to Solana. They work by locking assets on one chain and minting equivalent tokens on another. Bridges are one of the weakest points because they hold large amounts of funds and are complex. Bridges have accounted for a huge share of total losses in recent years:

- **Ronin Bridge (2022):** $625 million stolen through compromised private keys (related to Axie Infinity).
- **Wormhole (2022):** Approximately $325 million lost due to signature verification flaw in smart contract
- **Poly Network exploit (2021)**: Hackers stole over $600 million by exploiting weaknesses in the bridge system.

While total crypto and DeFi hack losses peaked significantly in 2022 at $3.8B after a massive multi-year surge, the subsequent drop and recent 2024 rebound to $2.2B indicate that ecosystem vulnerabilities remain a highly volatile and persistent threat.

Figure 3 — Crypto & DeFi Hack Losses by Year (USD Billions)
Source: Chainalysis Crypto Crime Reports



IX. BEST PRACTICES FOR IMPROVING BLOCKCHAIN AND USER SECURITY

- Developers can improve blockchain security by using strong access controls, role separation, multi-signature administration, extensive testing, code audits, formal verification where possible, and real-time monitoring for suspicious events.
- Multiple independent security audits before deployment are non-negotiable for protocols handling significant funds.
- Ethereum’s guidance recommends preparing emergency pause or upgrade mechanisms for crisis response.
- Multi-signature wallets also reduce the danger of a single compromised key.
- Use the Checks-Effects-Interactions pattern in smart contracts to prevent reentrancy vulnerabilities
- For users, the best defences are more basic but just as important: protect seed phrases, use hardware or well-secured wallets, verify websites carefully, avoid signing unknown transactions, and prefer platforms with transparent audits and stronger operational security.

X. FUTURE ADVANCEMENTS IN DECENTRALIZED CYBERSECURITY

The blockchain security space is changing rapidly. Here are a few new ways to lower risks in the future:

- AI-Powered Auditing — Using machine learning models to automatically scan smart contract code for known vulnerability patterns at scale.
- Zero-Knowledge Proofs (ZKPs) — Allows transactions to be cryptographically verified without exposing sensitive underlying data
- The future of decentralized cybersecurity will likely combine better coding practices with stronger infrastructure including safer smart contract frameworks, improved formal verification, more decentralized oracle systems, stronger bridge architectures, automated monitoring tools, and better wallet security design.
- Decentralized Identity (DID) — Giving users verifiable, self-owned digital identities that can help reduce impersonation and phishing.

XI. CONCLUSION

Decentralized technologies are powerful because they reduce dependence on central authorities, but they do not remove security problems. It creates new challenges on code, keys, protocol design and human layers. Addressing these challenges requires a layered defence strategy: rigorous auditing, formal verification and decentralised monitoring. The biggest lesson is that in decentralized systems, security is not only about encryption; it is also about secure programming, careful governance and strong user habits.

REFERENCES

[1] Ethereum Foundation. (2025). Smart Contract Security and Common Vulnerabilities. Ethereum Developers Guide

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[3] AI was used as a research assistance tool and structuring the report.